KIRAN KUMAR PRADHAN

AI & DATA SCIENCE | B.TECH UNDERGRADUATE AT NIT ROURKELA

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EDUCATION

National Institute of Technology Rourkela (NITR) Aug 2025 — Aug 2029

Bachelor of Technology (B.Tech) — Industrial Design CGPA: 8.53

EXPERIENCE

ML Developer Oct 2025 — Present

Machine Learning for Everyone (ML4E)

- Specialize in designing and training Machine Learning and Deep Learning models from scratch.
- Engineer robust model components and run hyperparameter tuning experiments to optimize model latency and accuracy.
- Collaborate with developers to explore new methods in model optimization and deployment pipeline architectures.

AI/ML Team Member Nov 2025 — Present

WEBWIZ, NIT Rourkela (WEBWIZ.NITR)

- Develop predictive Machine Learning models and clean/preprocess high-dimensionality structured and unstructured datasets.
- Generate visual insights and dashboards using Python, NumPy, Pandas, and Matplotlib.
- Contribute to club projects that bridge web-app technologies with intelligent machine learning backends.

KEY PROJECTS

Indian Constituent AI FastAPI, LlamaIndex, Groq API, HuggingFace

- Built a GovTech conversational assistant using Retrieval-Augmented Generation (RAG) to dynamically query the Indian Constitution.
- Implemented LlamaIndex with the Groq (Llama-3.3-70b-versatile) engine and HuggingFace embeddings for sub-second semantic search response times.
- Developed a lightweight and responsive web user interface connected to a FastAPI backend.

AI Career Path Recommender FastAPI, Groq API, Tavily Search, PyPDF

- Developed an agentic EdTech recommendation tool that parses resume files and maps tailored career roadmaps.
- Leveraged Llama 3 (via Groq API) combined with Tavily Search API to integrate real-time industry demand data.
- Structured backends using FastAPI, extracting content from multi-format PDFs and rendering interactive Jinja2 templates.

Book Recommendation System FastAPI, Pandas, Jinja2, Uvicorn

- Designed a collaborative and content-based recommendation engine for books using large Kaggle datasets.
- Managed data preprocessing pipelines and model serialization using Pandas and Pickle.
- Created a clean user interface with Jinja2 and served the system as a scalable FastAPI application.

Spam Classifier Website Scikit-Learn, NLTK, Streamlit, Python

- Created an NLP classification application using NLTK for text tokenization and TF-IDF Vectorizer for feature extraction.
- Trained a Multinomial Naive Bayes model to detect spam messages with high precision.
- Deployed the solution in a responsive, interactive Streamlit frontend for user accessibility.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML5, CSS3

Generative AI & LLMs: RAG, LlamaIndex, LangChain, Groq API, HuggingFace Embeddings

Machine Learning: Scikit-Learn, Naive Bayes, Cosine Similarity, Data Preprocessing

Data Science: Pandas, NumPy, Matplotlib, WordCloud, Data Visualization

Backend & Tools: FastAPI, Streamlit, Git, GitHub, Uvicorn, Jinja2 Templates

CERTIFICATIONS

- **Kaggle Dataset Creator Badge** — Indian Universities Directory
- **Python for Data Science** — Simplilearn Professional Certificate
- **A Primer to Deep Learning** — National Institute of Technology Rourkela
- **Intro to Machine Learning** — Kaggle Certificate of Completion